Are We Building Up or Drawing Down Human Capital?

Research note, September 2026. Projections generated with `npm run human-capital:trajectory` (default parameters; scenario and sensitivity runs noted where used); the 1925-2025 reconstruction with `scripts/human-capital-backcast.py`.

Question

Treat people times education as a capital asset: capitalize what it costs to rear and school each workforce entrant, and depreciate that cost over the years the entrant is expected to spend in the workforce. Then ask, region by region and year by year, whether the stock is growing or shrinking, and whether the world (and the United States in particular) is spending more or less on it than before.

The ledger is the `humanCapital` module documented in `docs/HUMAN_CAPITAL.md`: a Kendrick-style cost-based account at current replacement cost, four education bands, straight-line depreciation over an expected working life that nets out death, disability, domestic-role, and retirement exits. It is diagnostic: it reads demographics and GDP and feeds nothing back.

One accounting point first

The ledger prices every entrant as a multiple of the region's current GDP per capita, so its dollar stock rises with income even when the workforce it prices is shrinking. On the default path the world net stock goes from $321T in 2025 to $4,348T in 2100, a 13-fold rise that is almost entirely revaluation. To answer "growing or shrinking" the note deflates each region's stock by that region's own GDP-per-capita index (2025 = 1). The result is a constant-cost quantity: how many 2025-dollars of rearing and schooling are embodied in the people at work, at 2025 prices.

World

Year	Investment $T	Depreciation + write-offs $T	Net $T	Investment / GDP	Constant-cost net stock (2025 = 1)	Entrants (M)	Tertiary+ share of workforce
2025	21.4	20.7	+0.7	13.5%	1.00	119.2	24%
2030	24.3	23.0	+1.3	13.8%	1.05	119.6	27%
2040	32.6	31.8	+0.8	13.7%	1.16	118.8	37%
2050	44.6	46.2	-1.6	13.2%	1.24	116.3	45%
2060	59.8	65.8	-5.9	12.7%	1.27	112.8	52%
2075	89.3	103.7	-14.5	12.1%	1.26	106.7	59%
2100	214.6	253.4	-38.8	11.2%	1.19	95.7	59%

Three things stand out.

1. **The world is still building, but the build is ending.** The constant-cost stock grows about 1.1% a year through the early 2030s, half that by the 2040s, peaks in 2063 at 27% above 2025, and then drifts down about 0.2% a year to end 19% above 2025. Net investment at current cost turns negative in 2045; in constant-cost terms the turn comes in the 2060s because the revaluation of the opening stock is not a charge.
2. **What is being built is education, not headcount.** Entrants fall from 119M to 96M a year, but the tertiary-plus share of the in-service workforce goes from a quarter to three-fifths, and a tertiary entrant carries about 2.3 times the cost of a secondary one. Nearly all of the quantity growth to 2060 is composition; from the 2080s the tertiary headcount is itself falling.
3. **The spending share slips, not collapses.** Investment runs 13.5% of GDP now and 11.2% in 2100, because entrant cohorts shrink relative to GDP while per-entrant cost tracks income. Spending per entrant rises throughout; spending on entrants as a whole loses two points of GDP over the century.

By region

Constant-cost net stock index (2025 = 1), the peak year of that index, the first year own-cohort net investment is negative, the same including migration transfers, and the change in annual entrants over the century.

Region	2030	2050	2075	2100	Peak	Peak level	Net < 0	Net + migration < 0	Entrants 2100 / 2025
United States	1.11	1.37	1.46	1.49	2100	1.49	2035	2065	0.95
OECD ex-US	1.05	1.27	1.40	1.44	2100	1.44	2025	2086	1.11
China	0.94	0.76	0.58	0.42	2025	1.00	2025	2025	0.36
India + South Asia	1.15	1.62	1.52	1.27	2057	1.65	2061	2057	0.56
Latin America	1.04	1.16	1.07	0.88	2055	1.16	2069	2053	0.56
SE Asia + Pacific	1.04	1.25	1.21	1.00	2059	1.28	2072	2058	0.61
Russia + CIS	0.98	0.95	0.85	0.71	2025	1.00	2025	2025	0.59
MENA	1.24	1.98	2.09	1.97	2066	2.11	2056	2063	0.84
Sub-Saharan Africa	1.38	2.99	3.90	4.17	2100	4.17	never	never	1.26
World	1.05	1.24	1.26	1.19	2063	1.27	2045		0.80

The regions fall into four groups.

- **Drawing down from today: China, Russia + CIS.** China charges more depreciation than it capitalizes in every year of the run. Its entrant cohort falls by nearly two-thirds and its constant-cost stock by 58% by 2100, even though the rising college share means each entrant embodies more. In current dollars the Chinese stock still rises about 12-fold, which is the revaluation trap: a shrinking, better-educated workforce repriced at higher income looks like accumulation. Russia + CIS follows the same shape at a gentler slope.
- **Building to the 2050s, then drawing down: India + South Asia, Latin America, SE Asia + Pacific, MENA.** These are the demographic-dividend regions. India peaks in 2057 at 1.65 times its 2025 stock and gives back a quarter of the gain by 2100. MENA doubles and holds. India, Latin America, and SE Asia are

net exporters of trained people, and the outflow brings their turning points forward by 4 to 16 years; MENA is a small net receiver and its turn comes later once migration is counted.

- **Building throughout: Sub-Saharan Africa.** The only region whose entrant cohort is larger in 2100 than in 2025, and the only one whose own-cohort net investment never turns negative. Its stock quadruples, from a base that is 3% of the world total.
- **Building only through immigration: United States, OECD ex-US.** Both rich-region stocks rise through 2100, but own-cohort net investment is negative from 2025 in the OECD ex-US and from 2035 in the US. What keeps the stocks growing is the migration transfer: working-age arrivals booked at the destination's replacement cost. That transfer covers the own-cohort deficit until the mid-2060s (US) and the mid-2080s (OECD ex-US).

The United States

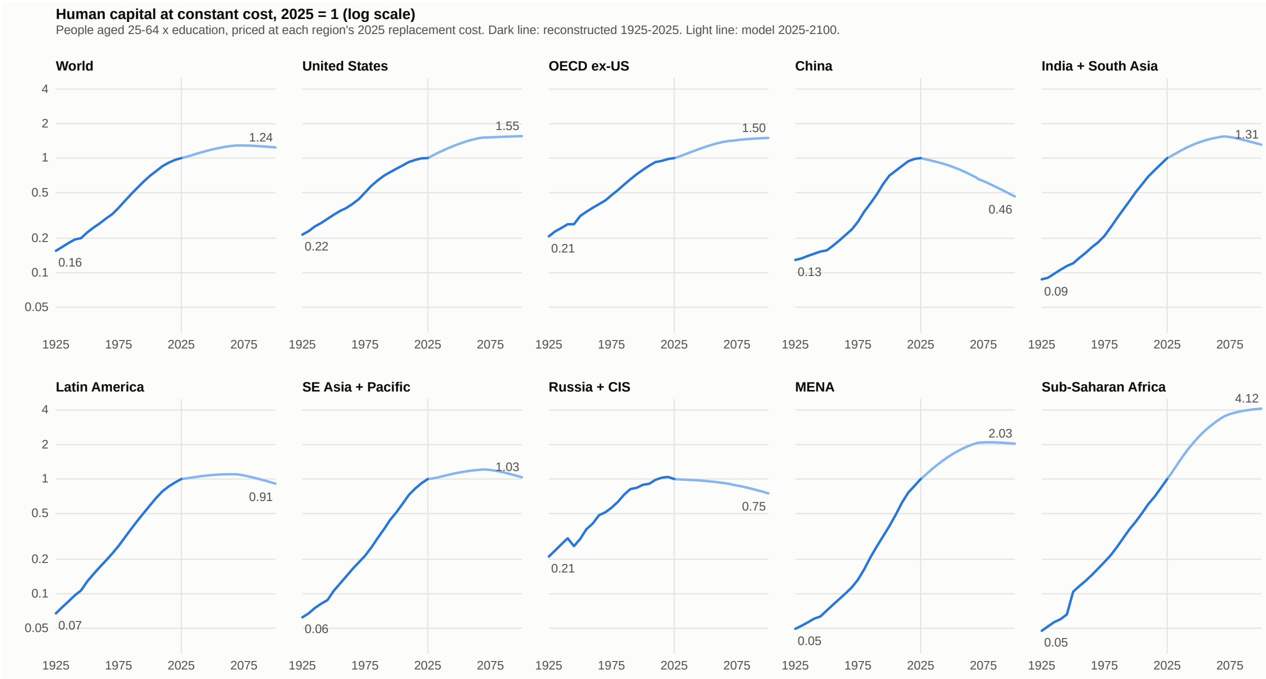
Year	Entrants (M)	Investment $T	Charge $T	Own-cohort net $T	Migration transfer $T	Net incl. migration $T	Investment / GDP
2025	4.1	3.98	3.58	+0.40	0.91	+1.31	16.6%
2035	4.0	4.64	4.69	-0.05	1.14	+1.09	15.6%
2050	4.0	6.89	8.01	-1.11	1.79	+0.68	14.6%
2075	3.9	13.02	16.52	-3.50	3.43	-0.07	13.6%
2100	3.9	32.92	41.92	-9.00	8.43	-0.58	12.9%

The US invests the largest share of GDP of any region in 2025 outside MENA and Sub-Saharan Africa, because its entrants are expensive (a 42% college share and the world's highest GDP per capita) rather than numerous. Its entrant cohort is flat at about 4.0M a year for the whole century, held up by a 1.4 fertility floor and net immigration of about 1.2M a year (UN WPP 2024 and CBO 2025 assumptions in `demographics.ts`). The own-cohort account turns negative in 2035, when the large cohorts now in mid-career start to retire faster than the flat entrant flow replaces them. Immigration is worth about a fifth of gross additions throughout, and it is what keeps the US stock growing. With migration switched off everywhere (`--set=demographics.migrationMultiplier=0`, which also removes migrants' children from future cohorts), US entrants fall by a third instead of 5%, the US constant-cost stock peaks in 2049 at 1.10 and ends the century at 0.91, and the OECD ex-US stock declines from 2025 to 0.79 by 2100. The US result is therefore an immigration-policy result at least as much as a demographic one.

The prior 100 years

The simulation starts in 2025 and cannot run backwards, so the century before it is reconstructed from observed data and priced with the same ledger. The stock is people aged 25-64 times their education (Lee and Lee's long-run attainment series, 1870-2040, and Barro-Lee by age group), valued at the ledger's replacement-cost multipliers; entrants are the population aged 20-24 divided by five, with the attainment mix of the cohort five years on. Costs are multiples of GDP per capita (Maddison), so investment and stock relative to GDP depend only on demographics and education, and the constant-cost stock holds each region's 2025 unit cost fixed.

Method and sources are in the script header; the reconstruction is gross (no age-based write-down), so the splice with the model uses the model's gross stock at constant cost.



Human capital at constant cost, 1925-2100, by region

Year	Population 25-64 (M)	Entrants (M/yr)	Tertiary share of 25-64	Entrants / population	Investment / GDP	Gross stock / GDP	Constant-cost stock (2025 = 1)
1925	769	35	1%	1.8%	11.7%	2.6	0.16
1950	1,042	44	1%	1.8%	12.3%	3.1	0.22
1975	1,575	72	3%	1.8%	14.7%	3.3	0.37
2000	2,778	104	8%	1.7%	14.5%	4.4	0.70
2010	3,342	124	10%	1.8%	15.8%	4.6	0.85
2025	4,083	125	10%	1.5%	14.1%	4.9	1.00

The world's stock of pre-workforce human capital grew about sixfold over the century, 1.9% a year: 1.7 points of that is more people of working age and 0.2 points is more education per person. Growth accelerated through each quarter-century to a peak of 2.6% a year in 1975-2000, then slowed to 1.4% in 2000-2025 and 0.9% in the last decade. The model's 1.1% for the late 2020s, falling to zero by the 2060s, is the continuation of that deceleration, not a break from it.

Constant-cost stock index (2025 = 1), reconstruction to 2025 and model after:

Region	1925	1950	1975	2000	2025	2050	2075	2100	1925-2025 (%/yr)	2025-2100 (%/yr)
United States	0.22	0.32	0.50	0.81	1.00	1.33	1.52	1.55	+1.5	+0.6
OECD ex-US	0.21	0.31	0.48	0.79	1.00	1.26	1.43	1.50	+1.6	+0.5
China	0.13	0.16	0.28	0.70	1.00	0.84	0.63	0.46	+2.0	-1.0
India + South Asia	0.09	0.12	0.21	0.51	1.00	1.37	1.53	1.31	+2.4	+0.4
Latin America	0.07	0.13	0.26	0.59	1.00	1.08	1.07	0.91	+2.7	-0.1
SE Asia + Pacific	0.06	0.11	0.21	0.51	1.00	1.14	1.20	1.03	+2.8	+0.0
Russia + CIS	0.21	0.30	0.56	0.89	1.00	0.96	0.88	0.75	+1.6	-0.4
MENA	0.05	0.07	0.13	0.39	1.00	1.64	2.09	2.03	+3.0	+0.9
Sub-Saharan Africa	0.05	0.10	0.19	0.43	1.00	2.33	3.68	4.12	+3.0	+1.9
World	0.16	0.22	0.37	0.70	1.00	1.19	1.29	1.24	+1.9	+0.3

Every region built human capital in every quarter-century of the last hundred years; the only setbacks in the series are Russia + CIS in 1940-1945 and again after 2020. The century ahead is the first in which any region draws down. China's reversal is the sharpest in the table: from the fastest builder of 1975-2000 (3.7% a year) to a 1% a year decline. Russia + CIS has already stalled, with its 2025 stock below 2015. The regions that grew fastest in the past century (MENA, Sub-Saharan Africa, SE Asia, Latin America, all around 3% a year) are the ones with the most building left, and only Sub-Saharan Africa keeps anything like its historical pace.

Are we spending more or less than we used to?

Investment in new entrants as a share of GDP, reconstructed:

Region	1925	1950	1975	2000	2010	2025
United States	15.6%	13.9%	18.6%	15.2%	16.4%	15.4%
OECD ex-US	10.9%	11.8%	13.5%	13.8%	13.5%	12.8%
China	9.6%	11.0%	15.6%	13.8%	18.4%	11.9%
India + South Asia	7.5%	9.6%	11.4%	14.6%	15.7%	16.6%
Latin America	10.5%	10.8%	12.8%	15.1%	15.2%	14.8%
SE Asia + Pacific	9.9%	10.6%	11.9%	15.2%	15.8%	15.9%
Russia + CIS	15.8%	17.9%	17.3%	17.4%	20.7%	13.2%
MENA	9.7%	9.3%	11.3%	15.7%	17.7%	16.8%
Sub-Saharan Africa	6.3%	9.7%	10.5%	12.7%	13.2%	15.8%
World	11.7%	12.3%	14.7%	14.5%	15.8%	14.1%

The world spends more of its output on new human capital than it did a century ago, and less than it did fifteen years ago. The share rose from under 12% of GDP in 1925 to a peak near 16% in 2010 as the education mix upgraded faster than cohorts shrank, and has since fallen back to 14% as the youth share of population dropped from 1.8% to 1.5%. Per entrant, the world spends 1.7 times its 1925 multiple of GDP per capita on rearing and schooling (5.7 to 9.6 times GDP per capita), and GDP per capita is itself far higher. The model's 13.5% for 2025 sits within a point of the reconstruction, and its slow decline to 11% by 2100 continues the post-2010 slide.

The United States:

Year	Entrants (M/yr)	Entrants / population	Entrant tertiary share	Workforce 25-64 (M)	Tertiary share of 25-64	Investment / GDP	Constant-cost stock (2025 = 1)
1925	2.09	1.87%	8%	54	5%	15.6%	0.22
1940	2.39	1.85%	10%	65	6%	16.4%	0.27
1950	2.35	1.53%	13%	77	8%	13.9%	0.32
1960	2.22	1.23%	16%	84	10%	11.8%	0.37
1970	3.47	1.67%	25%	92	14%	17.4%	0.44
1980	4.35	1.89%	32%	110	22%	20.5%	0.57
1990	3.86	1.52%	33%	131	28%	16.8%	0.70
2000	3.89	1.38%	33%	148	29%	15.2%	0.81
2010	4.46	1.44%	41%	166	33%	16.4%	0.92
2020	4.45	1.31%	40%	178	33%	15.0%	1.00
2025	4.61	1.34%	40%	178	33%	15.4%	1.00

The US series has one large hump: the baby boom entering the workforce with the first mass college cohorts, which took investment from under 12% of GDP in 1960 to over 20% in 1980. Since then the share has drifted between 15% and 17%, more education per entrant offsetting a falling youth share. The stock grew 1.5% a year over the century (1.2 points people, 0.3 education), but only 0.4% a year in the last decade, and the 25-64 population has been flat since 2020. The reconstruction reaches 2025 with the US stock already at a plateau; the model's 1.1% a year growth to 2050 therefore rests on its immigration assumption and on entrants' college share continuing to rise, as the previous section shows.

Outside data say the same. Total US education spending has held at about 5.4 to 5.6% of GDP in recent years, having peaked at 6.7% in 2009 (World Bank/UIS series), while the US birth cohort fell from 4.32M in 2007 to 3.63M in 2024 (NCHS). World government education spending was 4.16% of GDP in 2019 and 3.51% in 2023 (World Bank, from UNESCO UIS). The ledger's 14% is not comparable to those 4 to 6% education shares: it also counts rearing to entry age and students' foregone earnings. With rearing excluded the model's 2025 flow is 6.3% of GDP, close to the US total education share, and with foregone earnings excluded it is 11.3%.

Robustness

- **Energy and climate scenarios do not move the quantity path.** `ssp3-70` and `net-zero` produce identical constant-cost stocks, entrant flows, and regional peak years to the default run; only the dollar values differ, because those scenario files leave demographics untouched. The human-capital trajectory in this model is a demographic and education result, not an energy one.

- **Cost scope shifts levels and the world turning point, not the regional ordering.** Removing rearing raises every region's index (more of the cost is education, which is what is growing) and moves the world net-investment turn from 2045 to 2050; removing foregone earnings moves it to 2025. Peak years move by at most two years, except that Russia + CIS's flat stock becomes a slight rise to 2048 without rearing; the four regional groups are unchanged in both cases.

What the result does not say

- Cost is not value. A lifetime-income (Jorgenson-Fraumeni) account would be several times larger and would respond to wage premia this ledger ignores.
- Post-entry training, learning on the job, and obsolescence are outside the ledger; a shrinking pre-workforce stock is compatible with rising skill if on-the-job investment grows.
- Immigrants are booked at the destination's full replacement cost with no under-employment discount, so the rich-region migration transfers are an upper bound.
- The reconstruction is a gross, population-based stock (everyone aged 25-64, not the in-service workforce), with no age-based write-down and no migration accounting. Its regional groupings follow the model's nine regions, with all of Europe outside the CIS placed in OECD ex-US. Lee-Lee and Barro-Lee count China's completed tertiary attainment at about 3% of ages 25-64 in 2015 and 7% in 2025, well below the 2020 census figure of about 15% with junior college or above; China's stock growth since 2000 and its 2025 education level are therefore understated, which makes its projected decline, if anything, start from a higher base. Before 1950 the covered countries are scaled to regional population totals, which assumes uncovered countries had the covered average attainment.
- Demographics follow the UN WPP 2024 low variant, so entrant cohorts outside Sub-Saharan Africa shrink faster than a medium-variant path would give; the medium variant would delay every peak year but not remove the ordering.

Sources

- World Bank, *Government expenditure on education, total (% of GDP)* (SE.XPD.TOTL.GD.ZS), from UNESCO Institute for Statistics: world 4.16% (2019), 3.51% (2023); United States series and the 2009 peak.
- NCHS, *Births in the United States, 2024* (Data Brief 535) and *National Vital Statistics Reports* 74(1): 4,316,233 births in 2007, 3,628,934 in 2024, total fertility rate 1.63.
- Lee, J.-W. and Lee, H. (2016). "Human Capital in the Long Run." *Journal of Development Economics* 122: attainment by level for ages 15-24 and 25-64, 1870-2010, with projections to 2040 (OUP long-run files); Barro, R. and Lee, J.-W. (2013), v3 dataset by 10-year age group, 1950-2015.
- UN DESA, *World Population Prospects 2024*, population by 5-year age group, medium variant; Maddison Project Database 2023 (GDP per capita, 2011$) and Our World in Data population series (Gapminder/HYDE/UN).
- `docs/HUMAN_CAPITAL.md` for the ledger's method, calibration, and prior art (Kendrick 1976; Eisner 1985; Mallatt 2026; Eurostat duration of working life).